An Adaptive Pseudo-Labeled Sample Generation Approach to Unsupervised Multiclass Change Detection in Bitemporal Remote Sensing Images

Sicong Liu, *Senior Member, IEEE*, Kecheng Du, Lorenzo Bruzzone, *Fellow, IEEE*, Qian Du, *Fellow, IEEE*, Francesca Bovolo, *Senior Member, IEEE*, and Xiaohua Tong, *Fellow, IEEE*

***Abstract*—Multiclass change detection (CD) in multitemporal multispectral/hyperspectral remote sensing images is a significant yet challenging task, particularly when prior knowledge and ground reference data are not available. This article proposes a novel adaptive pseudo-labeled sample generation (APSG) approach to address this challenge, which bridges the gap between data-driven unsupervised CD and supervised CD methodologies. In particular, the spectral difference image is first projected into a 2-D polar domain using an adaptive spectral change vector representation (ASCVR). Subsequently, a novel method for generating multiple pseudo-labeled candidate sample selection regions is developed to automatically analyze the statistical distribution and characteristics of change and no-change classes in bitemporal images, thereby producing high-confidence pseudo-labeled samples. Finally, a sequential pseudo-labeled sample selection and generation strategy incorporated into the CD process is designed to enable robust multiclass CD. Multiclass CD results are further enhanced by incorporating spatial–spectral information. The effectiveness of the proposed method has been validated on three publicly available bitemporal remote sensing datasets, outperforming the traditional CD approaches and the deep learning (DL)-based unsupervised multiclass CD methods in terms of the improvement of overall accuracy (OA) (1%–5%). The generated pseudo-labeled samples represent intrinsic change clusters and their feature distributions within the dataset, thus resulting in a data-driven approach. These pseudo-labeled samples serve as critical inputs for both machine learning (ML)- and DL-based CD methods utilizing multispectral/hyperspectral data at medium-to-low spatial resolution. This approach enables reliable multiclass CD across diverse land-cover scenarios without requiring prior information.**

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Sicong Liu, Kecheng Du, and Xiaohua Tong are with the College of Surveying and Geoinformatics, Tongji University, Shanghai 200092, China (e-mail: sicong.liu@tongji.edu.cn; kecheng_du@tongji.edu.cn; xhtong@tongji.edu.cn).

Lorenzo Bruzzone is with the Department of Information Engineering and Computer Science, University of Trento, 38123 Trento, Italy (e-mail: lorenzo.bruzzone@unitn.it).

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA (e-mail: du@ece.msstate.edu).

Francesca Bovolo is with the Fondazione Bruno Kessler, Center for Information and Communication Technology, 38123 Trento, Italy (e-mail: bovolo@fbk.eu).

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I. Introduction

With the advancement of the new generation of remote sensing satellites, the volume of acquired remote sensing data is growing exponentially [1], [2]. Consequently, a wealth of multitemporal images is available in data archives, making it possible to detect changes on the Earth's surface over large areas, either within a short time interval or over a long evolutionary period. Change detection (CD) techniques quantitatively analyze and identify these land-cover changes using bitemporal remote sensing image data covering the same region [3], [4]. Over recent decades, CD techniques have become indispensable in diverse remote sensing application domains, including forestry, disaster monitoring, urban management, agriculture, and wetland monitoring [5], [6], [7], [8], [9]. However, effectively mining and extracting the rich temporal change information associated with natural evolution and human activities remains highly challenging. This difficulty arises primarily from the lack of ground reference data in operational scenarios, especially when analyzing completely unknown areas without any available prior knowledge. Therefore, developing advanced unsupervised data-driven CD approaches remains a significant and valuable research focus [10].

Depending on the availability of prior information, CD methods are broadly categorized into supervised and unsupervised techniques [3], [11].

The former requires a portion of labeled ground reference data to train a classifier. This training can utilize either traditional machine learning (ML) methods, such as support vector machine (SVM), random forests (RFs) [12], [13], or deep learning (DL) techniques, such as residual networks (ResNet), convolutional neural network (CNN), U-Net, and long short-term memory network (LSTM) [14], [15], [16], [17], [18]. By taking advantages from the available prior information, supervised CD methods usually achieve higher accuracy than unsupervised approaches [19], [20], [21]. However, collecting comprehensive ground reference data for bitemporal images is inherently labor-intensive, time-consuming, and sometimes infeasible, particularly in complex scenarios such as disaster

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damage assessment or urban environment monitoring. In this context, several studies focus on generating a small number of pseudo-labeled samples as a training set, by using approaches such as combining compressed change vector analysis (C^2VA) [10] with iterative reweighted multivariate CD (IR-MAD) [22] or generative adversarial network (GAN) [23]. However, most existing methods are specifically designed for binary CD, which aims only to locate changed pixels. The more challenging task of multiclass CD, which requires not only detecting changed pixels but also discriminating their change types, remains less studied. Consequently, current methods have limited applicability to complex multiclass CD scenarios.

In contrast, unsupervised CD methods operate without requiring training samples or prior knowledge of the study area, showing enhanced applicability across diverse scenarios [1], [24], [25]. In the literature, many unsupervised CD techniques have been proposed, including the change vector analysis (CVA) [26], IR-MAD [27], the temporal-principal component analysis (TPCA) [28], and the slow feature analysis (SFA) [29]. To extract features from bitemporal images that emphasize the changes, clustering or thresholding algorithms are subsequently applied to these features to extract specific change information [30]. Among these techniques, data-driven cluster analysis, particularly in relation to CVA-based methods, has emerged as a prominent approach to unsupervised CD. Originally proposed for analyzing multiple change types using different band combinations in bitemporal remote sensing images [26], [31], the standard CVA is limited by its ability to detect changes occurring only in specific spectral bands. To overcome this limitation, C^2VA was developed, compressing spectral change vectors (SCVs) from all bands and projecting them into a 2-D polar coordinate domain to represent diverse changes [32]. A sequential SCV analysis (S^2CVA) method was designed to further optimize the compressed representation by iteratively calculating an adaptive reference vector during SCV projection [33]. Recent research has explored enhancements through spectral–spatial techniques [10], band expansion [34], and DL adaptations, such as DCVA [35] and DSFA [36]. Moreover, in order to remove the pretrained models in DL adaptations, an attempt has been conducted by integrating optimization modeling into untrained networks and achieved superior performances in unsupervised hyperspectral image CD [37].

A number of studies have investigated the generation of pseudo-labeled samples for training supervised and unsupervised CD methods, aiming to bridge the gap between these approaches and enable automated CD. There are three categories of methods presented in the literature for generating pseudo-labeled samples, including clustering-based, CVA-based, and thresholding-based methods. These methods share a core objective: evaluating unlabeled samples' confidence levels to select high-confidence instances as pseudo-labeled samples [38]. For example, Gao et al. [39] employed Gabor wavelets and fuzzy c-means (FCMs) algorithms for preclassification, selecting pixels with high confidence as changed or unchanged samples. In [40], the Otsu thresholding algorithm is implemented on CVA-derived data to generate a binary CD map, with pseudo-labeled samples refined through a neighborhood-based criterion that reduces noise and misclassification pixels by assessing local label consistency. In [41], an unsupervised CD method was proposed based on CVA and selective Bayesian thresholding to derive a pseudo-training set. Li et al. [38] acquire initial pseudo-samples through a multiscale preclassification approach, subsequently undergoing iterative refinement through a Bayesian network to ultimately generate a larger and more accurate pseudo-sample dataset. Binary CD was subsequently performed in the original bitemporal feature space. Although these methods have shown effectiveness, they focus solely on the binary CD problem by generating two-class (i.e., changed and unchanged) samples, with limited attention given to the more complex multiclass CD scenario. In our previous work, we introduced a semi-supervised multiclass CD (S^2MCD) framework [42]. This framework generates a multiclass pseudo-training set using S^2CVA and achieves promising performance when applied to bitemporal hyperspectral images. Pseudo-labeled sample generation methods based on difference image analysis and thresholding are also widely adopted. In [43], local binary patterns and CVA are combined to extract object-specific features, while the expectation maximization (EM) algorithm was employed to compute thresholds. Pseudo-labeled samples are then generated based on the resulting threshold segmentation. Shao et al. [44] applied a threshold selection strategy based on Bayesian decision theory to the difference image, identifying high-confidence regions of change and unchange samples, and leveraging a large uncertainty interval for pseudo-labeled sample generation. Wu et al. [45] designed a deep siamese KPCA convolutional mapping network to extract change features. These features were projected into the polar coordinate domain for clustering and segmentation, resulting in unsupervised multiclass CD maps. In addition, several studies have employed DL frameworks to generate or optimize pseudo-samples, thereby achieving multiclass CD results [46], [47]. In [48], an adaptive semi-supervised learning method is proposed to improve pseudo-sample accuracy and optimize the training process. However, these methods still require a small amount of prior information as initial input, which cannot achieve a fully unsupervised process.

Although the aforementioned techniques have demonstrated effectiveness in various applications using real remote sensing datasets, several open issues remain that require further investigation. First, existing methods often rely on thresholding or clustering to distinguish between change and no-change classes and then generate pseudo-labeled samples through simple confidence calculations. However, the underlying distribution characteristics of each change class and the no-change class within temporal image pairs remain insufficiently explored. This limitation may negatively affect the CD process and contribute to both omission and commission errors. Second, most current studies focus on generating binary pseudo-labeled samples, whereas only limited efforts address the more complex task of multiclass CD. Third, further challenges arise in the multiclass pseudo-labeled sample generation process, including sample imbalance and poor model

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generalization ability, both of which require thorough analysis and effective solutions.

To address the aforementioned challenges, a novel adaptive pseudo-labeled sample generation (APSG) approach is proposed to enable automatic multiclass CD. The key innovations and contributions of the proposed approach are as follows.

1) It provides a simple yet effective solution to the problem of pseudo-labeled sample generation. By analyzing the statistical distribution of change and no-change samples in a 2-D compressed representation domain, APSG automatically identifies multiple candidate regions for pseudo-labeled sample selection, each corresponding to different confidence levels. As a result, this data-driven framework operates in a fully automatic and unsupervised manner, exhibiting wide applicability across multiclass CD tasks involving various spatial and spectral resolutions.

2) It employs a sequential strategy to sequentially generate the pseudo-labeled sample training set and achieve multiclass CD results, thereby bridging the methodological gap between unsupervised change mining and supervised change discrimination. Through a coarse-to-fine process evolving from binary to multiclass CD, the pseudo-labeled sample set is progressively updated, enhancing sample diversity while preserving confidence levels.

3) It jointly integrates spectral–spatial change information into both the pseudo-labeled sample generation and the multiclass CD process. This integration enhances the representativeness and stability of the pseudo-labeled sample training set while significantly reducing errors caused by one-shot discrimination and random sample generation.

Experimental results obtained on one multispectral and two hyperspectral CD datasets confirmed the effectiveness of the proposed approach by comparing it with state-of-the-art CD methods.

The rest of this article is organized as follows. Section II reviews the related works. Details of the proposed APSG approach are presented in Section III. Experimental results and analysis based on three real benchmark datasets are presented in Section IV. Finally, Section V summarizes the main conclusions.

II. Related Work

Let X_1 and X_2 be a pair of coregistered multispectral or hyperspectral remote sensing images acquired at times t_1 and t_2, respectively, and composed of B spectral channels. SCVs in the difference image X_D can be calculated by the vector subtraction of the bitemporal images

$$X_D = X_2 - X_1. \tag{1}$$

The considered multiclass CD problem can be formulated as follows. Let $\Omega = \{\Omega_{\mathrm{NC}}, \Omega_C\}$ be the set of all classes in X_D, where Ω_{NC} is the set of no-change class (C_n) and $\Omega_C = \{\Omega_C^1, \Omega_C^2, \ldots, \Omega_C^K\}$ represents the set of K change classes ($C_1, C_2, \ldots, C_K$).

A. Adaptive SCV Representation

Adaptive SCV representation (ASCVR) was designed as a part of S^{2}CVA method to solve a multiclass CD problem in bitemporal hyperspectral images by following a hierarchical analysis [10]. Two change variables are employed to build the compressed 2-D polar domain, i.e., magnitude ρ and direction θ [33]. In particular, ρ is the Euclidean compression of SCVs that reflects the spectral brightness of the change vectors (i.e., it is the magnitude of the SCVs), while θ is defined in terms of the spectral angle distance that reflects the shape difference of SCVs related to different change classes [49]. The two change variables are defined as

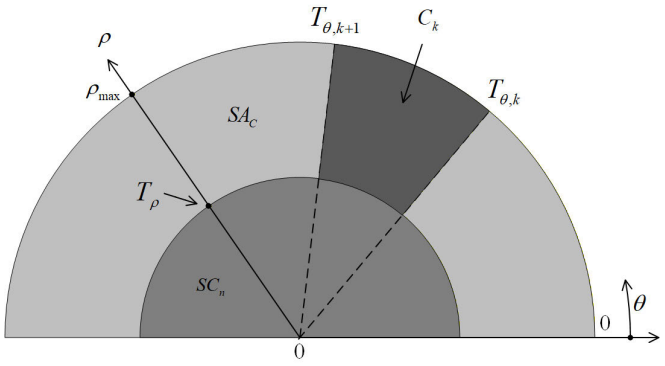


Fig. 1. Multiclass change representation in the 2-D polar domain based on S^{2}CVA.

$$\rho = \sqrt{\sum_{b=1}^{B} \left(X_D^b\right)^2} \tag{2}$$

$$\theta = \arccos\left[\left(\sum_{b=1}^{B} X_D^b r^b\right) \Big/ \sqrt{\sum_{b=1}^{B} \left(X_D^b\right)^2 \sum_{b=1}^{B} \left(r^b\right)^2}\right] \tag{3}$$

where X_D^b is the bth ($b = 1,\ldots, B$) component of X_D and r^b is the bth component of an adaptive reference vector $\boldsymbol{r}$. S^{2}CVA is based on the first eigenvector of the covariance matrix A of X_D [33]

$$A = \mathrm{cov}\,(X_D) = E\left[(X_D - E\,[X_D])\,(X_D - E\,[X_D])^T\right] \tag{4}$$

where $E[X_D]$ is the expectation of X_D. The eigendecomposition of A can be represented as

$$A \cdot V = V \cdot W \tag{5}$$

where W is a diagonal matrix with the eigenvalues being sorted in descending order (i.e., $\lambda^1 > \lambda^2 > \cdots > \lambda^B$) in the diagonal. V is the matrix of eigenvectors (i.e., $V = [v^1, v^2, \ldots, v^B]$). The reference vector $\boldsymbol{r}$ is defined as the first eigenvector v^1 corresponding to the largest eigenvalue λ^1, i.e., $r = v^1$. In this way, the variance of the measurement is maximized while maintaining the change information of different categories, which effectively improves the change representation.

Accordingly, the compressed 2-D polar domain D is defined by ρ and θ as follows [32], [33]:

$$D = \left\{\rho \in \left[0, \rho_{\max}\right] \text{ and } \theta \in [0, \pi]\right\} \tag{6}$$

where $\rho_{\max}$ denotes the maximum value of ρ. All SCVs can be represented in the 2-D semicircular polar coordinate domain D (see Fig. 1). In this domain, the unchanged SCVs have low-magnitude values, so they are generally distributed in SC$_n$. The

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changed SCVs are distributed in SA_C, which exhibit higher magnitude values. The regions of SC_n and SA_C are defined as:

$$SC_n = \{\rho \in [0, T_\rho) \text{ and } \theta \in [0, \pi]\} \quad (7)$$

$$SA_C = \{\rho \in [T_\rho, \rho_{\max}] \text{ and } \theta \in [0, \pi]\}. \quad (8)$$

In SA_C, the spectral difference of SCVs associated with different change classes is reflected in the value of the direction θ. Thus, the separation between change (Ω_C) and no-change (Ω_{NC}) SCVs can be accomplished by applying a threshold T_ρ to the magnitude ρ, and multiclass changes ($\Omega_C^1, \ldots, \Omega_C^K$) can be identified by setting multiple thresholds $T_{\theta,k}$ ($k = 1,..., K-1$) in SA_C along the direction θ.

B. Extended Multi-AP

Extended multiattribute profiles (EMAPs) are a type of shallow spatial–spectral features that have been widely used in CD [50], [51]. Attribute profiles (APs), which serve as an extended form of morphological profiles (MPs), form the basis for constructing EMAP features. In detail, given a single grayscale image α, APs are obtained by processing it according to a criterion μ, with m attribute thickening (ϕ^μ) operators and m attribute thinning (γ^μ) operators

$$AP(\alpha) = \{\phi_m^\mu(\alpha), \varphi_{m-1}^\mu(\alpha), \ldots, \phi_1^\mu(\alpha), \alpha, \gamma_1^\mu(\alpha), \ldots, \gamma_{m-1}^\mu(\alpha), \gamma_m^\mu(\alpha)\}. \quad (9)$$

Considering a multispectral or hyperspectral image with B channels, extended APs (EAPs) can be derived based on APs. The EAP can be expressed as

$$EAP = \{AP(g_1), AP(g_2), \ldots, AP(g_B)\} \quad (10)$$

where $g_1, g_2,..., g_B$ are the image bands. Details of selected attributes are the same in [52]. Multiple different EAPs can be computed by considering different attributes to construct the final EMAP as

$$EMAP = \{EAP_a, EAP_d, EAP_t, EAP_s\} \quad (11)$$

where a is the area attribute, d is the length of the diagonal of the box bounding the region attribute, t is the moment of inertia attribute, and s is the standard deviation of the gray-level pixel values attribute. The final EMAP is the combination of all EAPs extracted using the above four parameters.

III. Proposed Adaptive Pseudo-Sample Generation Approach

By taking advantage of statistical analysis of multiclass SCVs in the polar domain, multiple pseudo-labeled sample selection regions are adaptively segmented, allowing for the generation of high-confidence pseudo-labeled samples. This process facilitates a robust unsupervised CD method that does not rely on reference data or prior knowledge, effectively bridging the gap between data-driven unsupervised CD and supervised CD methodologies. The proposed APSG approach comprises three main steps: 1) pseudo-labeled sample selection region generation based on ASCVR; 2) sequential pseudo-labeled sample generation; and 3) multiclass CD.

The overall workflow of the proposed approach is illustrated in Fig. 2. Section III provide detailed explanations of each step.

A. Pseudo-Labeled Sample Selection Region Generation Based on ASCVR

This step is designed to define regions that facilitate the extraction of pseudo-labeled sample sets in subsequent steps. This is achieved using the ASCVR, where all compressed SCVs exhibit different clusters corresponding to unchanged pixels and different kinds of changed pixels within the compressed 2-D polar domain. Precise segmentation of the ASCVR domain is critical for producing effective pseudo-labeled sample sets and ensuring accurate multiclass CD.

In our previous work, we employed magnitude thresholding to distinguish between change and no-change samples, thereby obtaining a binary CD result [42]. However, real-world CD scenarios are often more complex, as multiclass changes tend to form distinct clusters. In particular, subtle changes typically exhibit lower change magnitude values compared to those of major changes in hyperspectral images [53], making a single threshold on the change magnitude prone to errors. In this work, we reverse the sequence of operations. Specifically, we first partition the semicircular polar domain into multiple sectors based on predefined thresholds along variable θ, with each sector representing a specific type of change and the related part of the no-change pixels. Subsequently, we estimate individual thresholds along variable ρ within each sector to obtain an accurate binary separation between each change class and related samples of the no-change class in the polar domain.

Let T_0 denote the threshold value calculated via the EM algorithm applied to the ρ values of all pixels [32]. It is used to provide a rough binary CD result, dividing the entire sample set Ω into the two subsets Ω_C and Ω_{NC} in the polar domain. For the Ω_C subset, only the direction θ is analyzed to identify multiple change classes. This is done by using the K-means method to segment the entire Ω_C into K sectors. The boundaries of each sector on θ are denoted by A_{i-1} and A_i ($i = 1, 2,..., K$, where when $i = 1$ then $A_{i-1} = 0$, and when $i = K$ then $A_i = \pi$). Note that the cluster number K can be estimated as proposed in [10]. Finally, for each sector Ω^i, magnitude segmentation is conduced again by using the EM method to calculate T_1^i, aiming to optimize the binary CD and divide samples within this region into Ω_C^i and Ω_{NC}^i (represented as Ω_ω^i and $\omega \in \{C, NC\}$, where the magnitude of Ω_ω^i is denoted by ρ_ω^i).

In theory, under proper assumptions, the magnitudes of the change and no-change samples within each sector should ideally follow the Rice and Rayleigh mixture distribution that can be approximated with a Gaussian mixture distribution. The two classes can be separated by T_1^i [Fig. 3(a)]. However, as illustrated in Fig. 3(b), the long-tail effect caused by an uneven sample distribution, commonly observed in real datasets, makes threshold determination for T_1^i highly challenging.

To address the limitation, we introduce a sample distribution mode judgment criterion to assess whether a sector's sample

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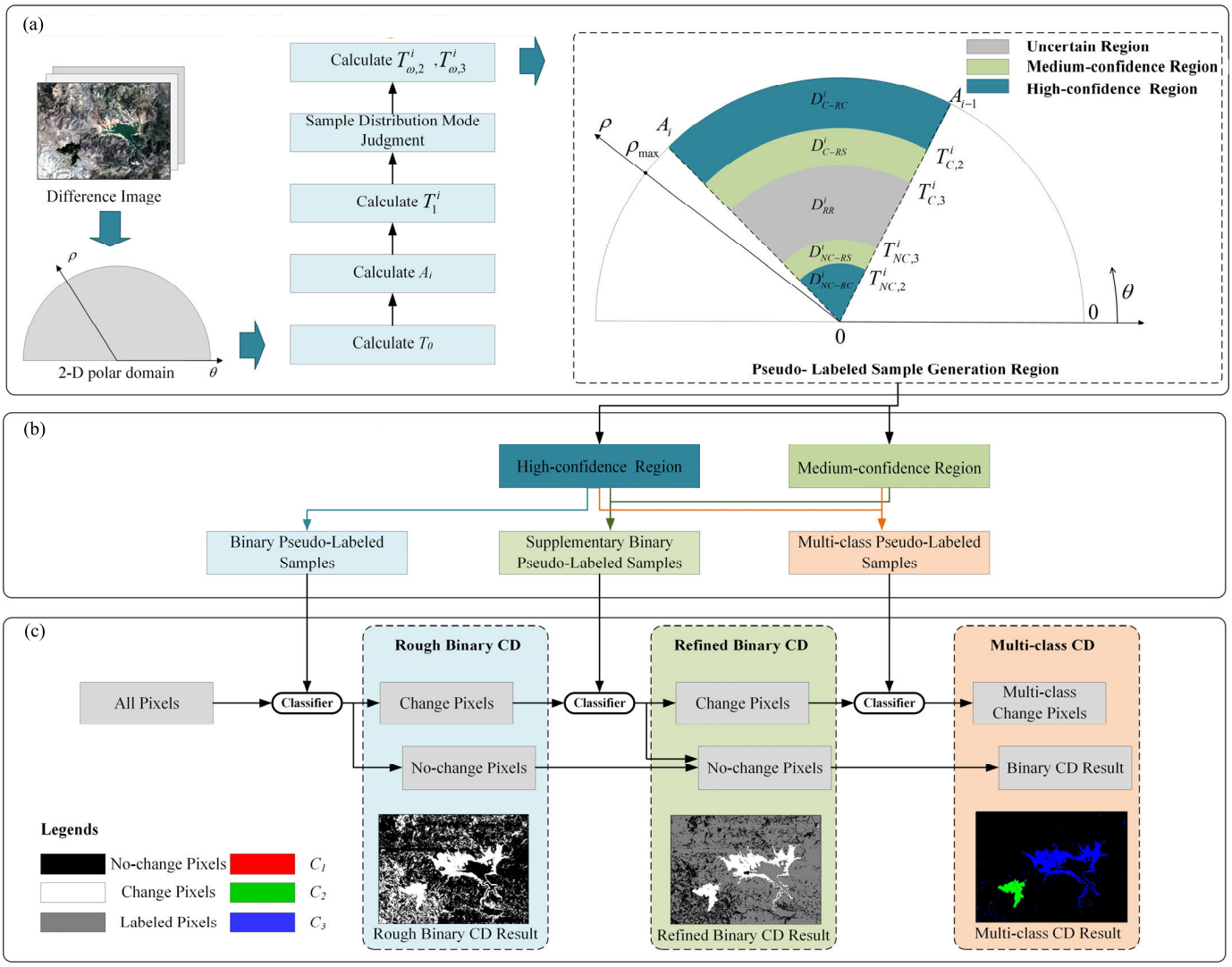


Fig. 2. Block diagram of the proposed approach. (a) Pseudo-labeled sample selection region generation. (b) Sequential pseudo-labeled sample generation. (c) Multiclass change detection.

distribution is conformed to a Gaussian mixture model on ρ^i_ω. Subsequently, based on the validity of this judgment on Ω^i_ω, different strategies are applied for threshold calculation and sample selection. The criterion is implemented through frequency histogram analysis based on ρ^i_ω, with detailed methodological descriptions provided in the following. We approximate the magnitude value of the cluster center on Ω^i_ω with H^i_ω computed as

$$H^i_\omega = \left(H^i_{\omega,\mathrm{me}} + H^i_{\omega,\mathrm{mo}}\right)/2 \tag{12}$$

where $H^i_{\omega,\mathrm{me}}$ and $H^i_{\omega,\mathrm{mo}}$ represent the median and mode of frequency histogram statistics on Ω^i_ω, respectively.

As defined in (13) and (14) (where $\rho^i_{\omega,\max}$ and $\rho^i_{\omega,\min}$ are the maximum and minimum values of ρ^i_ω, respectively), a threshold S^i_ω is calculated on ρ^i_ω to detect the long-tail effect, setting as 1/3 distance from T^i_1 (namely, $\rho^i_{C,\min}$ for change class and $\rho^i_{\mathrm{NC},\max}$ for no-change class) within the magnitude interval

$$S^i_C = \left(\rho^i_{C,\max} - \rho^i_{C,\min}\right)/3 + \rho^i_{C,\min} \tag{13}$$

$$S^i_{\mathrm{NC}} = \frac{2}{3}\left(\rho^i_{\mathrm{NC},\max} - \rho^i_{\mathrm{NC},\min}\right) + \rho^i_{\mathrm{NC},\min}. \tag{14}$$

The sample distributions of Ω^i_C and Ω^i_{NC} are considered as long-tail effects when $S^i_C > H^i_\omega$ ($\omega = C$) and $S^i_{\mathrm{NC}} < H^i_\omega$ ($\omega = \mathrm{NC}$), respectively. Long-tail distribution patterns can be detected through positional offsets of real clusters within magnitude intervals.

According to the Gaussian mixture model, no-change and change samples form distinct clusters. In regions near the boundary between these clusters, samples are often intermixed, leading to ambiguous classifications, which involve extremely low-confidence levels during the pseudo-labeled sample generation process. In contrast, samples located closer to the cluster centers or at intensity extremes tend to exhibit higher confidence. However, high-confidence samples often lack diversity and do not adequately represent the boundary regions. Therefore, relying solely on high-confidence samples for the pseudo-labeled sample training sets may result in poor training labels, particularly in CD, where the boundaries of changed areas become blurred. Thus, by considering both pseudo-labeled sample confidence and information diversity, $T^i_{\omega,2}$ and $T^i_{\omega,3}$ (i.e., $T^i_{C,2}, T^i_{\mathrm{NC},2}, T^i_{C,3}$, and $T^i_{\mathrm{NC},3}$) are

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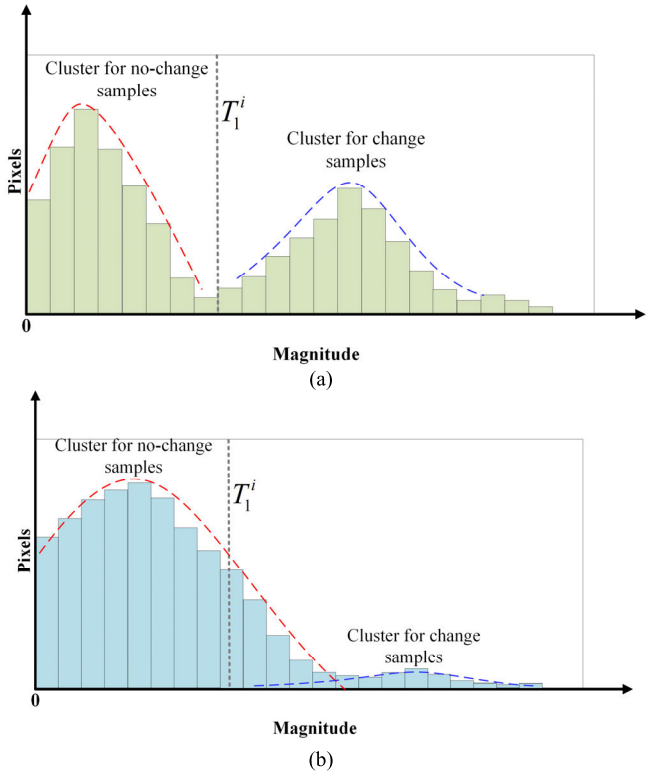


Fig. 3. Frequency histogram of samples within a sector. T_1^i is the threshold calculated by the EM algorithm. (a) Change and no-change sample clusters conform to a Gaussian mixture distribution and (b) long-tail effect caused by insufficient change sample size.

derived from ρ_ω^i to separate each sector into five regions: the no-change high-confidence pseudo-labeled sample selection region $D_{\mathrm{NC-RC}}^i$, the no-change medium-confidence pseudo-labeled sample selection region $D_{\mathrm{NC-RS}}^i$, the uncertain region D_{RR}^i, the change medium-confidence pseudo-labeled sample selection region $D_{C-\mathrm{RS}}^i$, and the change high-confidence pseudo-labeled sample selection region $D_{C-\mathrm{RC}}^i$. These regions will be further analyzed using a specific strategy to generate a set of complete and reliable pseudo-labeled samples.

The frequency histogram of ρ distribution serves as the basis for subsequent calculations of $T_{\omega,2}^i$ and $T_{\omega,3}^i$. $T_{\omega,2}^i$ can be obtained by (15) and (16) based on ρ_ω^i of Ω_ω^i

$$T_\omega^i = \arg\min\left(\left|f(t) - F_\omega^i\right|\right) \tag{15}$$

$$f(t) = \frac{\left|t - \rho_{\omega,e}^i\right| N_t}{LN} \tag{16}$$

where t represents the possible threshold values requiring determination. L is the range of ρ_ω^i, which can be calculated by subtracting $\rho_{\omega,\max}^i$ and $\rho_{\omega,\min}^i$. N is the number of samples in Ω_ω^i. $|t - \rho_{\omega,e}^i|$ represents the distance between the threshold value t and the magnitude extreme value $\rho_{\omega,e}^i$ ($\rho_{\omega,e}^i = \rho_{C,\max}^i$ when $\omega = C$, and $\rho_{\omega,e}^i = \rho_{\mathrm{NC},\min}^i$ when $\omega = \mathrm{NC}$). The smaller the distance, the higher the sample confidence. N_t denotes the number of samples with magnitudes falling within the interval $[t, \rho_{\omega,e}^i]$. Accordingly, the function $f(t)$ essentially represents the product of two ratios: the sample count ratio and the magnitude range ratio between Ω_ω^i and the region partitioned by threshold t. The value of $T_{\omega,2}^i$ can then be derived from

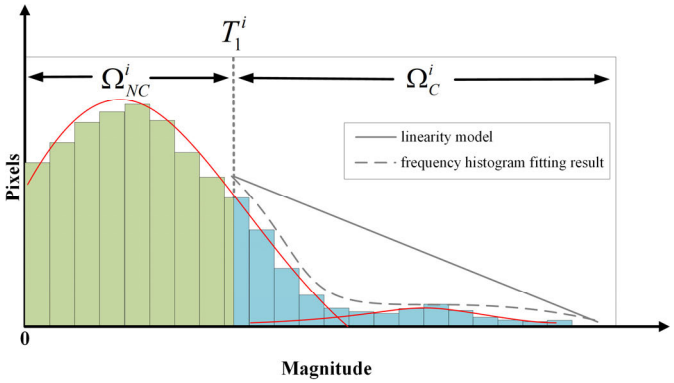


Fig. 4. Linear model approximation of the long-tail effect (Ω_C^i) in the sample frequency histogram along the magnitude direction.

(15), where F_ω^i is employed to quantify histogram features and modulate the confidence interval range.

For samples with an approximately Gaussian distribution, F_C^i and F_{NC}^i can be simply set as 1/4 and 1/2, respectively. However, when the distribution of Ω_ω^i exhibits a long-tail effect in frequency histograms, setting F_ω^i as a constant may easily introduce errors in T_ω^i. Thus, we propose an algorithm to adaptively adjust F_ω^i based on the specific sample distribution. F_ω^i can be computed as:

$$F_\omega^i = \frac{\chi\left(P_C\right)^2}{\left(P_\omega^i\right)^2} \tag{17}$$

$$P_\omega^i = \frac{P_{\omega,1}^i + P_{\omega,2}^i}{4P_{\omega,1}^i P_{\omega,2}^i}. \tag{18}$$

Here, the long-tail effect is approximated by using a linear model, as illustrated in Fig. 4. In this approximation [see (17)], $P_C = 1/3$ is defined as the confidence boundary, and $\chi = 1.25$ represents the empirical parameter deviation between the linear model and actual long-tail effect. The adaptive parameter P_ω^i, calculated based on the empirical distribution [see (18)], reflects the clustering behavior of the data.

In greater detail, the no-change sample set Ω_{NC}^i with long-tail distribution, $P_{\mathrm{NC},1}^i$ and $P_{\mathrm{NC},2}^i$ can be calculated as

$$\begin{cases} P_{\mathrm{NC},1}^i = \left(\tau_2^i - \tau_1^i\right)/L \\ P_{\mathrm{NC},2}^i = \left(\rho_{\mathrm{NC},\max}^i - \tau_1^i\right)/L. \end{cases} \tag{19}$$

For Ω_C^i, $P_{C,1}^i$ and $P_{C,2}^i$ can be calculated as

$$\begin{cases} P_{C,1}^i = \left(\tau_2^i - \tau_1^i\right)/L \\ P_{C,2}^i = \left(\tau_2^i - \rho_{C,\min}^i\right)/L \end{cases} \tag{20}$$

where τ_1^i and τ_2^i ($\tau_1^i < \tau_2^i$) are parameters used to quantify the primary cluster centers of samples within the dataset, which can be obtained as

$$\begin{cases} L_s = \arg\min\left(\tau_2^i - \tau_1^i\right) \\ N_\tau = 0.75N. \end{cases} \tag{21}$$

In other words, the number of samples N_τ within the interval $[\tau_1^i, \tau_2^i]$ accounts for 75% of Ω_ω^i, with the distance between τ_1^i and τ_2^i (i.e., L_s) being minimized.

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Once $T^i_{\omega,2}$ is determined, Ω^i_ω is divided into two parts. Samples falling within the intervals $[T^i_{C,2}, \rho^i_{C,\max}]$ or $[\rho^i_{\mathrm{NC},\min}, T^i_{\mathrm{NC},2}]$ are classified as high-confidence samples. Those in the intermediate confidence range (i.e., $[T^i_{\omega,2}, T^i_1]$) exhibit relatively lower confidence levels compared to the rest, but still contain potentially important discriminatory information for CD. Accordingly, $T^i_{\omega,3}$ is calculated based on the remaining samples using the same abovedescribed method used for $T^i_{\omega,2}$. Finally, each sector can be represented by five regions, as illustrated in Fig. 2 as follows.

1) No-Change High-Confidence Pseudo-Labeled Sample Selection Region:

$$D^i_{\mathrm{NC-RC}} = \left\{\rho \in \left[0, T^i_{\mathrm{NC},2}\right], \theta \in [A_{i-1}, A_i]\right\}.$$

2) No-Change Medium-Confidence Pseudo-Labeled Sample Selection Region:

$$D^i_{\mathrm{NC-RS}} = \left\{\rho \in \left[T^i_{\mathrm{NC},2}, T^i_{\mathrm{NC},3}\right], \theta \in [A_{i-1}, A_i]\right\}.$$

3) Uncertain Region:

$$D^i_{\mathrm{RR}} = \left\{\rho \in \left[T^i_{\mathrm{NC},3}, T^i_{C,3}\right], \theta \in [A_{i-1}, A_i]\right\}.$$

4) Change Medium-Confidence Pseudo-Labeled Sample Selection Region:

$$D^i_{C-\mathrm{RS}} = \left\{\rho \in \left[T^i_{C,2}, \rho^i_{C,\max}\right], \theta \in [A_{i-1}, A_i]\right\}.$$

5) Change High-Confidence Pseudo-Labeled Sample Selection Region:

$$D^i_{C-\mathrm{RC}} = \left\{\rho \in \left[T^i_{C,3}, T^i_{C,2}\right], \theta \in [A_{i-1}, A_i]\right\}.$$

The calculation process for $T^i_{\omega,2}$ and $T^i_{\omega,3}$ is summarized in Algorithm 1.

Algorithm 1 Adaptive Threshold Calculation.

Input: Mag^i (the magnitude of Ω^i_ω), T^i_1 and $Flag^i_\omega$
Output: $T^i_{\omega,2}$ and $T^i_{\omega,3}$

```
for i = 1 to K do
    if ω = C then
        if Flag^i_ω = 1 then
            F^i_C = 1/4
            Calculate T^i_{ω,2} by (15)-(16) on Ω^i_C
            Calculate T^i_{ω,3} by (15)-(16) on [T^i_1, T^i_{C,2}]
            else {Flag^i_ω = 0}
            Calculate F^i_C by (17)-(21) on Ω^i_C
            Calculate T^i_{ω,2} by (15)-(16) on Ω^i_C
            n is number of samples on [T^i_{C,2}, ρ^i_{C,max}]
            Select the top n maximum magnitude on
            [T^i_1, T^i_{C,2}], T^i_{ω,3} is the nth maximum magnitude
            value
        end if
    else {ω = NC}
        if Flag^i_ω = 1 then
            F^i_NC = 1/2
            Calculate T^i_{ω,2} by (15)-(16) on Ω^i_NC
            Calculate T^i_{ω,3} by (15)-(16) on [T^i_{NC,2}, T^i_1, ]
        else {Flag^i_ω = 0}
            Calculate F^i_NC by (17)-(21) on Ω^i_NC
            Calculate T^l_{ω,2} by (15)-(16) on Ω^i_NC
            n is number of samples on [0, T^i_{NC,2}]
            Select the top n minimum magnitude on
            [T^i_{NC,2}, T^i_1], T^i_{ω,3} is the nth mini- mum magnitude
            value
end if
    end if
end for
```

B. Sequential Pseudo-Labeled Sample Generation

To minimize the negative impact of pseudo-labeled sample errors on supervised classifiers, we propose a sequential pseudo-labeled sample generation strategy incorporated into the CD process (i.e., rough binary CD, refined binary CD, and multiclass CD) using the pseudo-labeled training set derived from distinct confidence regions. This strategy directly influences CD results at each stage. Fig. 2 illustrates the entire pseudo-labeled sample generation process.

In the first stage, 10% of the samples are randomly selected from the $D^i_{\mathrm{NC-RC}}$ and $D^i_{C-\mathrm{RC}}$ regions (i = 1, 2,..., K) as pseudo-labeled samples representing the no-change and change classes, respectively. Based on these pseudo-labeled samples, a rough binary CD result is obtained using a simple SVM classifier. Note that the pseudo-labeled sample set generated by the proposed approach can be input into any classifier. The pseudo-labeled samples generation strategy adopts a conservative approach, operating only within high-confidence regions during this phase. As a result, the change class in the rough binary CD results typically exhibits a low missed-alarm rate but a high false-alarm rate. The second stage is designed to remove the numerous false detections present in the change class.

The refined binary CD process focuses exclusively on change class regions identified in the rough binary CD output, while excluding no-change regions in the polar domain. To enhance the ability to distinguish changes, pseudo-labeled samples are generated from both high-confidence regions ($D^i_{C-\mathrm{RC}}$ and $D^i_{\mathrm{NC-RC}}$) and medium-confidence regions ($D^i_{C-\mathrm{RS}}$ and $D^i_{\mathrm{NC-RS}}$). Sector-wise processing is applied to minimize false detections caused by interclass variations within change samples. For each sector, 10% of the samples from $D^i_{\mathrm{NC-RC}}$ and 5% from $D^i_{\mathrm{NC-RS}}$ are randomly selected to form the no-change training set, with the remaining samples being used for refined binary CD. The change class training set is constructed in a similar way. To mitigate class imbalance, the sampling ratio is dynamically adjusted for underrepresented change classes. For classes yielding fewer than 20 pseudo-samples, the generation proportion is increased to 50%, ensuring sufficient training samples for these classes. Binary CD is then performed using an SVM classifier within each sector, and the final binary CD results are obtained by aggregating the change class regions across all sectors after removing no-change samples.

Following the exclusion of no-change class samples identified through rough and refined binary CD, the remaining samples are designated as change samples for classification into K distinct change classes. To minimize interclass confusion along the θ direction, the top and bottom 10% of θ

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values within each sector are removed. Subsequently, 10% of the remaining samples are randomly selected to serve as multiclass pseudo-labeled training samples.

Beyond mitigating pseudo-labeled sample errors through the aforementioned strategies, using spectral–spatial features as classifier inputs further enhances the accuracy of both the pseudo-labeled samples and CD results. Since pseudo-labeled sample generation directly influences rough and refined binary CD results, the selection of input features significantly affects this process. Prior analysis demonstrates that integrating spectral–spatial features, such as EMAPs, as classifier input effectively reduces pseudo-labeled sample error rates and improves CD accuracy [10], [34]. These features reduce the influence of erroneous samples by incorporating spatial-contextual information, which preferentially retains isolated minor changes while more effectively suppressing background noise. EMAPs are particularly well suited for this framework due to their ability to represent multiscale and multimodal spatial information.

C. Multiclass CD

Multiclass CD results can be derived simply by classifying the changed samples obtained in the binary CD map derived at the previous step. Utilizing pseudo-labeled training samples for each change class, as described previously, all change samples are categorized into K classes using the multiclass SVM classifier. The classification accuracy is enhanced by incorporating the EMAPs spectral–spatial features into the classifier input. Finally, multiclass CD results are generated by integrating binary CD outcomes with the classified change samples.

IV. Experimental Results

Experimental validation was performed on three remote sensing benchmark datasets (one with multispectral and two with hyperspectral images). The effectiveness of the proposed approach, denoted as APSG, was evaluated both qualitatively and quantitatively by comparing it with five state-of-the-art methods: standard C^2VA [32], improved C^2VA (IC^2VA) [54], S^2CVA [33], a semi-supervised multiclass CD method (S^2MCD) [42], and an unsupervised multiclass CD method based on DL model (kernel principal component analysis convolutional mapping network, KPCA-MNet) [45]. A variant incorporating EMAPs as input features for the SVM classifier is denoted as APSG_EMAP (SVM). In addition, we conducted experiments using the VGG16 network model [55] as a classifier to validate the effectiveness of the pseudo-samples generated by APSG in DL models, denoted as APSG (VGG16). Due to inherent randomness, performance metrics for S^2MCD, APSG (SVM), APSG_EMAP (SVM), and APSG (VGG16) were averaged over 20 trials. To ensure comparable training sample sizes, the pseudo-labeled sample selection ratio for S^2MCD was adjusted to 10%. Comprehensive evaluation employed five metrics: overall accuracy (OA), Kappa coefficient (Kappa), overall error (OE), F_1 score, and computational time [56]. Specifically, the confusion matrix was calculated according to the reference image, and other four values were further computed, i.e., true positive (TP), true negative (TN), false positive (FP), and false negative (FN),

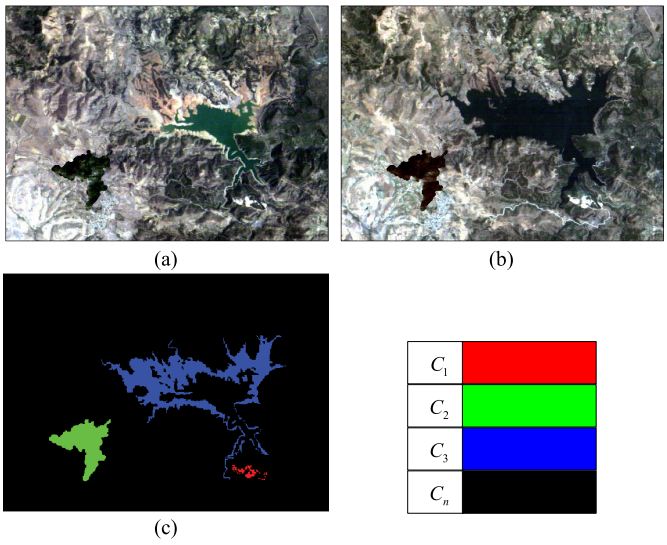


Fig. 5. RGB composites of the bitemporal Landsat TM images (bands 1–3) of Sardinia Island (Italy) acquired in (a) 1995 (X_1) and (b) 1996 (X_2). (c) Change reference map.

which are defined as the number of pixels correctly classified as change and unchanged classes, and the number of pixels misclassified as change and unchanged classes. OA is the proportion of all correctly classified pixels to all pixels, which can be calculated as

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{22}$$

OE denotes the total number of misclassified pixels

$$\mathrm{OE} = \mathrm{FP} + \mathrm{FN}. \tag{23}$$

To mitigate the impact of class imbalance on accuracy assessment, Kappa is commonly employed to reflect overall consistency, calculated as

$$\begin{cases} \mathrm{Kappa} = \dfrac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}} \\ \mathrm{PRE} = \dfrac{(\mathrm{TP} + \mathrm{FP}) \cdot \mathrm{RC} + (\mathrm{TN} + \mathrm{FN}) \cdot \mathrm{RU}}{(\mathrm{FP} + \mathrm{FN} + \mathrm{TP} + \mathrm{TN})^2} \end{cases} \tag{24}$$

where RC and RU indicate the number of real changed and unchanged pixels in the reference image, respectively. In addition, the F_1 score is employed to consider both precision and recall of the model. It can be calculated as

$$F_1 = 2 \cdot \frac{P \cdot R}{P + R} \tag{25}$$

where $P = (\mathrm{TP}/(\mathrm{TP} + \mathrm{FP}))$ and $R = (\mathrm{TP}/(\mathrm{TP} + \mathrm{FN}))$ represent the precision and recall rate, respectively.

A. Dataset 1: Sardinia

Sardinia dataset includes a pair of real bitemporal Landsat-5 TM images acquired in September 1995 (X_1) and July 1996 (X_2), over Sardinia Island (Italy). The image size is 412 × 300 pixels with six spectral bands. There are three land-cover changes in this dataset, including an enlargement of an open quarry (i.e., C_1), a fire affect area (i.e., C_2), and an enlargement of lake surface due to rising water levels of Lake Mulargia (i.e., C_3). More details on this dataset can be found in [32]. Fig. 5(a) and (b) shows RGB composites of X_1 and X_2,

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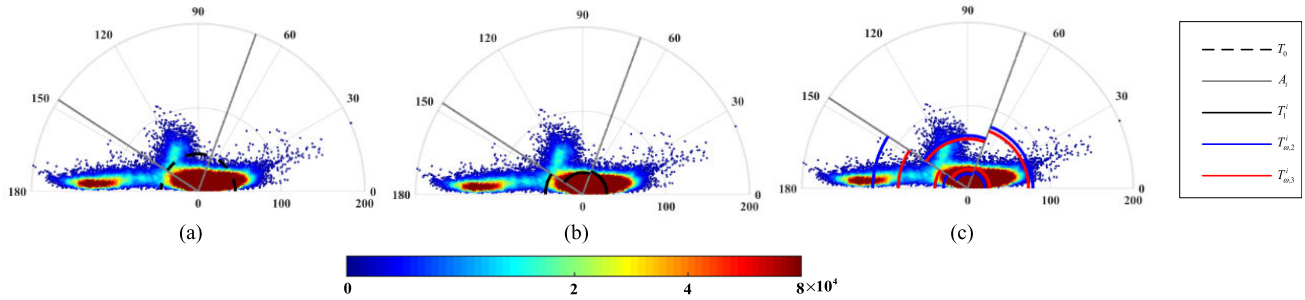


Fig. 6. Pseudo-labeled sample selection region generation result (Sardinia dataset). (a) Sector regions corresponding to each change classes, (b) change and no-change sample region separation in each sector, and (c) final pseudo-labeled sample selection regions.

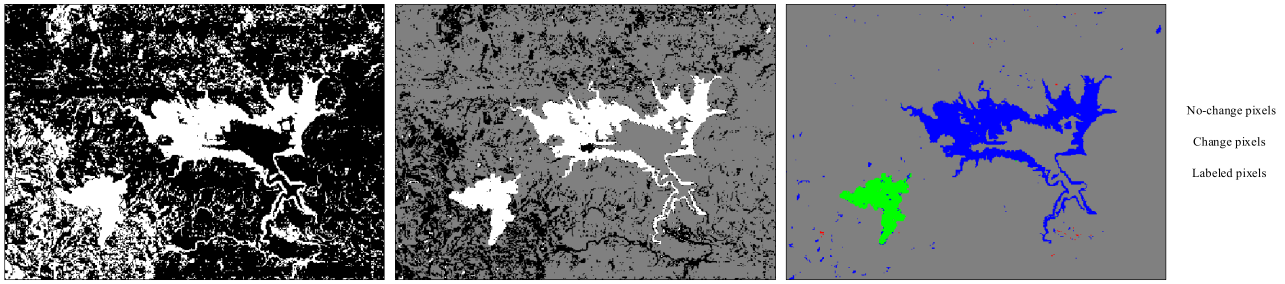


Fig. 7. CD maps obtained in three sequential stages. (a) Rough binary CD, (b) refined binary CD, and (c) multiclass CD (Sardinia dataset).

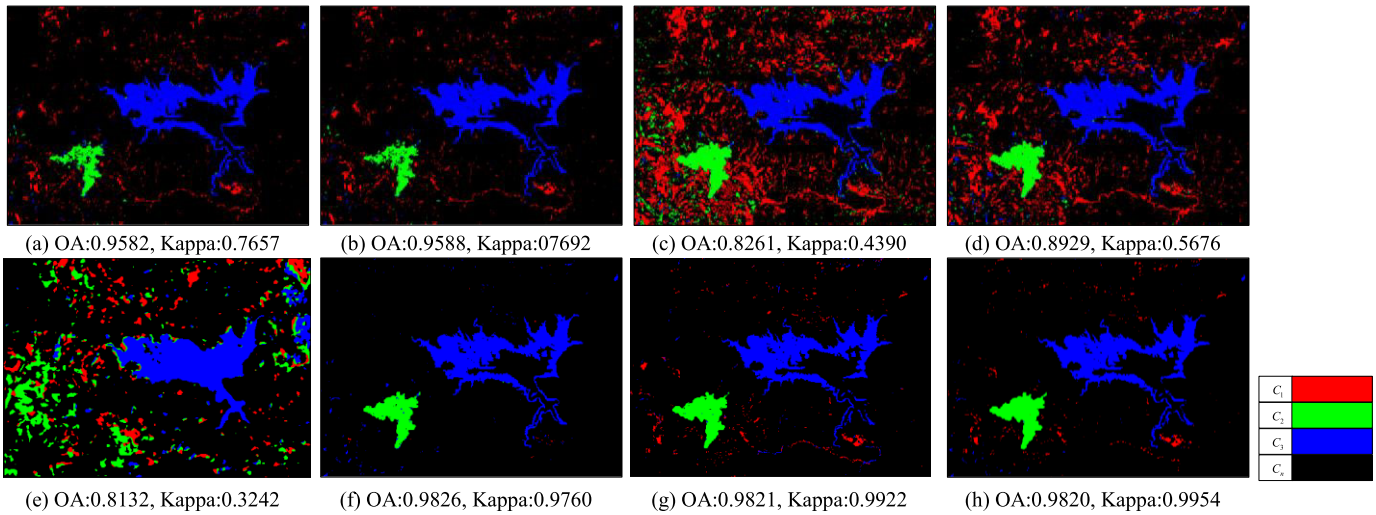


Fig. 8. Multiclass CD result on Sardinia dataset. (a) C^2VA, (b) S^2CVA, (c) IC^2VA, (d) S^2MCD, (e) KPCA-MNet, (f) APSG (SVM), (g) APSG (VGG16), and (h) APSG_EMAP (SVM).

respectively. Fig. 5(c) shows the change reference map of the dataset, where three change classes are labeled as red (i.e., C_1, 214 pixels), green (i.e., C_2, 2414 pixels), and blue (i.e., C_3, 7480 pixels), and the no-change class is labeled as black (i.e., C_n, 113 492 pixels).

Fig. 6 illustrates the pseudo-labeled sample selection region generation result in the 2-D semicircular polar domain based on the defined thresholds. Three distinct change clusters can be observed and represented in the 2-D scatter plot, leading to the predefined cluster count $K = 3$. The polar domain angle interval (from 0 to 180) is divided into three sector regions corresponding to change classes C_1–C_3. As illustrated in Fig. 6(b), the thresholds T_1^1 and T_1^2 calculated by the EM algorithm fails to accurately segment change clusters, exhibiting bias toward the no-change cluster. Consequently, segmentation results for change regions C_1 and C_2 are considered inaccurate due to the long-tail effect following sample distribution mode judgment.

Thus, the final thresholds in Fig. 6(c) are adopted to generate pseudo-labeled training samples for no-change class and each change class. By incorporating Gaussian mixture modeling and long-tail effect strategies, medium-confidence pseudo-labeled sample generation regions are defined in the polar coordinate domain using $T_{C,2}^i$ and $T_{C,3}^i$ ($i = 1, 2$), effectively reducing mislabeled samples.

To further illustrate the sequential strategy employed throughout the CD process (from binary to multiclass), a representative trial is presented to show the result of each stage

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TABLE I
RESULTS OBTAINED BY DIFFERENT METHODS ON SARDINIA DATASET

Method	OA	Kappa	OE (pixels)	F_1	Time(s)
C^2VA	0.9582	0.7657	5,166	0.7811	15.57
S^2CVA	0.9588	0.7692	5,089	0.7851	**15.50**
IC^2VA	0.8261	0.4390	21,463	0.4822	24.23
S^2MCD	0.8883±0.0025	0.5578±0.0056	13,811±304	0.5913±0.0051	139.98±10.93
KPCA-MNet	0.8132	0.3242	23091	0.3468	53.46
APSG(SVM)	0.9664±0.0280	0.8758±0.1499	4,727±4,366	0.8167±0.1330	171.64±1.01
APSG(VGG16)	0.9732±0.0117	0.9571±0.0477	2,898±539	0.8380±0.0371	322.41±5.73
APSG_EMAP(SVM)	**0.9776±0.0033**	**0.9885±0.0128**	**2,510±345**	**0.8862±0.0135**	175.60±3.67

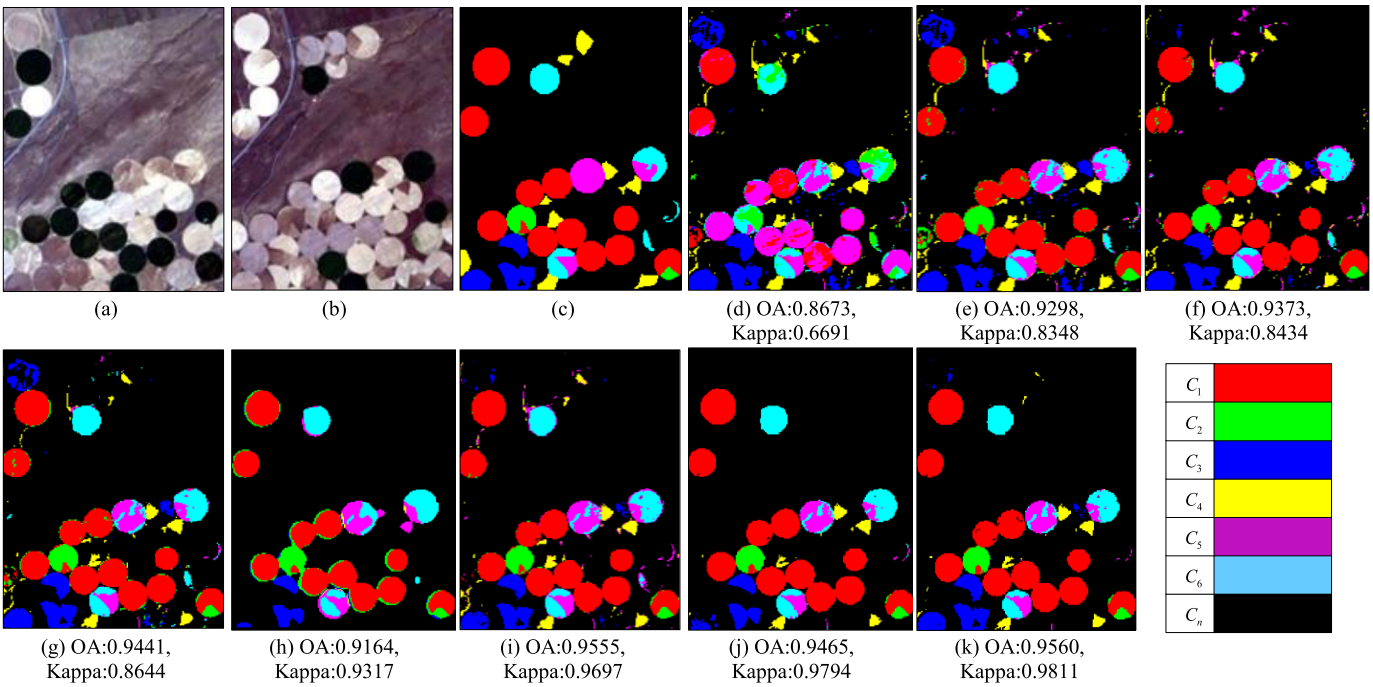


Fig. 9. False color composite (R: 650.67 nm, G: 548.92 nm, and B: 447.17 nm) of Umatilla dataset (U.S.). (a) Images acquired in 2004 ($\boldsymbol{X}_1$) and (b) images acquired in 2007 ($\boldsymbol{X}_2$). (c) Reference map. (d)–(i) Multiclass CD result obtained by different methods: (d) C^2VA, (e) S^2CVA, (f) ICVA, (g) S^2MCD, (h) KPCA-MNet, and (i) APSG (SVM). (j) and (k) APSG_EMAP (SVM).

(see Fig. 7). The rough binary CD result [Fig. 7(a)] exhibits excessive background noise, where numerous no-change pixels are erroneously classified as changes. Because pseudo-labeled samples are generated solely from high-confidence regions (i.e., $D^i_{C-\mathrm{RC}}$ and $D^i_{\mathrm{NC}-\mathrm{RC}}$), many change and no-change samples remain inadequately separated in the polar domain, leading to misclassification in this stage. To address this issue, additional pseudo-labeled training samples are generated from both high- and medium-confidence regions (i.e., $D^i_{C-\mathrm{RS}}$ and $D^i_{\mathrm{NC}-\mathrm{RS}}$). The refined binary CD result [Fig. 7(b)] demonstrates that these samples provide essential change information to the training set, enabling sector-wise binary CD to effectively reduce false detection rates in the second stage. Finally, a precise multiclass CD result is derived by classifying only the change pixels identified in the refined binary CD result [Fig. 7(c)], excluding no-change regions. Fig. 8 presents the multiclass CD results of the proposed approach and reference methods, with corresponding evaluation metrics provided in Table I, which confirms the effectiveness of the proposed APSG approach. OA value increased from 0.9588 (S^2CVA) to 0.9664 (APSG), and further increased to 0.9776 (APSG_EMAP), with about half wrongly classified OE pixels reduced (i.e., OE was reduced from 5089 pixels in S^2CVA to 2510 $\pm$ 345 pixels in APSG_EMAP). In multiple CD results, the change class C_3 is well separated via single-threshold magnitude analysis in IC^2VA and S^2MCD [Fig. 8(c) and (d)]. Conversely, C_1 and C_2 exhibit higher error rates due to severe class imbalance. Although S^2MCD incorporates supervisory information, its pseudo-labeled sample selection strategy fails to improve C_1 detection accuracy due to persistent class imbalance. In contrast, S^2CVA and C^2VA [Fig. 8(a) and (b)] enhance cluster boundary identification in polar coordinates using manual discrimination criteria. However, separation based on compressed change vector representation is insensitive to subtle spectral changes, leading to residual noise. Although KPCA-MNet extracts spatial–spectral features via deep networks, the low spatial resolution of the dataset results in poor performance for classes C_1 and C_2. Compared to the baselines Fig. 8(a)–(e), the proposed approach [see Fig. 8(f)] significantly reduces background noise caused by inaccurate threshold for change C_1 in the polar domain, though some subtle change [e.g., C_1 in Fig. 8(f)]

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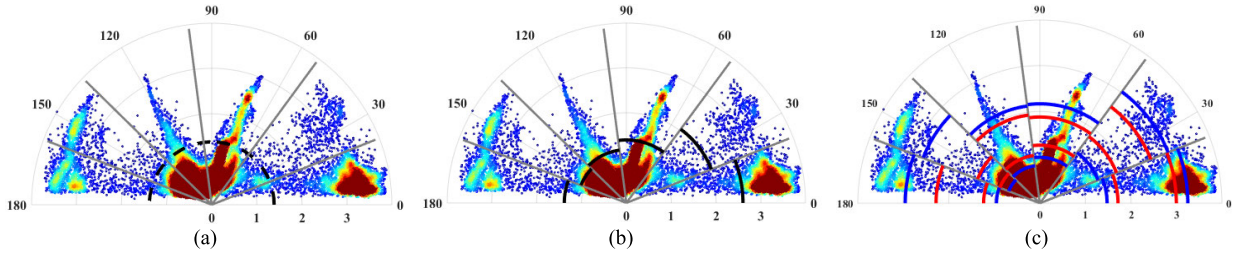


Fig. 10. Pseudo-labeled sample selection region generation results (Umatilla dataset). (a) Sector regions corresponding to each change classes, (b) change and no-change sample region separation in each sector, and (c) final pseudo-labeled sample selection regions.

TABLE II
RESULTS OBTAINED BY DIFFERENT METHODS ON UMATILLA DATASET

Method	OA	Kappa	OE (pixels)	F_1	Time (s)
C^2VA	0.8673	0.6691	10,087	0.4512	9.76
S^2CVA	0.9298	0.8348	3,941	0.8366	**8.96**
IC^2VA	0.9373	0.8434	2,555	0.8622	12.22
S^2MCD	0.9392±0.0023	0.8539±0.0050	2,461±94	0.8736±0.0045	215.56±54.66
KPCA-MNet	0.9164	0.9317	3385	0.9594	67.91
APSG(SVM)	0.9432±0.0128	0.9541±0.0674	2,521±294	0.8605±0.0152	143.70±13.37
APSG(VGG16)	0.9420±0.0019	0.9569±0.0323	2,817±155	**0.9640±0.0133**	297.13±4.05
APSG_EMAP(SVM)	**0.9492±0.0038**	**0.9666±0.0312**	**2,201±185**	0.8757±0.0119	136.09±3.72

may be ignored. This limitation is mitigated by integrating spatial–spectral features (i.e., EMAPs) as input to the SVM classifier. For Sardinia dataset, EMAPs are derived from the first three principal components, with parameters being $a = 500$, $d = 200$, $t = 0.5$, and $s = 2.5$. As demonstrated in Fig. 8(h) and Table I, APSG_EMAP achieves superior multiclass CD accuracy and stability, particularly for class C_1. Furthermore, the APSG (VGG16) effectively detects the C_1 change by extracting spatial information, demonstrating the effectiveness of pseudo-samples it generates. The proposed method outperforms the baselines on Sardinia dataset, achieving the highest accuracy metrics through EMAP-based feature extraction (i.e., OA = 0.9776 ± 0.0033, Kappa = 0.9885 ± 0.0128, and $F_1 = 0.8862 \pm 0.0135$).

B. Dataset 2: Umatilla

Umatilla dataset comprises a pair of real bitemporal hyperspectral remote sensing images acquired by the Hyperion sensor aboard the EO-1 satellite on May 1, 2004 ($\boldsymbol{X}_1$) and May 8, 2007 ($\boldsymbol{X}_2$). The study area covers irrigated agricultural land in Umatilla County, Oregon, U.S., spanning 180×225 pixels. Following a series of preprocessing steps, 159 spectral bands are retained out of 242 original bands. Land-cover changes in this study area are mainly driven by transitions among bare land, crops, soil, and water. The imbalanced distributions of change classes coupled with their subtle spectral characteristics make the unsupervised CD more challenging. Fig. 9(a)–(c) shows false-color composites of $\boldsymbol{X}_1$, $\boldsymbol{X}_2$, and the change reference map, which includes six change classes (i.e., C_1–C_6) and the no-change class (i.e., C_n) in this area.

Fig. 10 illustrates the pseudo-labeled sample generation process. The semicircular polar coordinate domain (from 0° to 180°) is divided into six sectors using thresholds A_i, corresponding to classes C_1–C_6 in the counterclockwise order. As shown in Fig. 10(b), the thresholds T_1^i for C_1, C_3, and C_4 (i.e., T_1^1, T_1^3, and T_1^4) lack sufficient power to distinguish between change and no-change clusters. Consequently, after sample distribution analysis, the change regions for C_3 and C_4 and the no-change region for C_1 are identified, as exhibiting long-tail distributions.

The numerical experimental results are provided in Table II, and the multiclass CD maps are displayed in Fig. 9(d)–(k). Agricultural land-cover changes in this dataset typically show weak spectral features, which become ambiguous when compressed into the 2-D polar coordinate system. As illustrated in Fig. 9(i)–(k), background noise is effectively suppressed. For change classes with high intraclass variation, incorporating a small proportion of pseudo-labeled samples from medium-confidence regions proves effective. Furthermore, for classes with imbalanced sample sizes, adding underrepresented change classes to the training model can reduce FP rates. Consequently, the proposed methods achieve higher OA and Kappa values [i.e., OA = 0.9432 ± 0.0128 and Kappa = 0.9541 ± 0.0674 for APSG (SVM) and OA = 0.9420 ± 0.0019 and Kappa = 0.9569 ± 0.0323 for APSG (VGG16)], and the use of EMAPs features further enhances CD performance (i.e., OA = 0.9666 ± 0.0312 and Kappa = 0.9666 ± 0.0312). For Umatilla dataset, EMAPs are derived from the first four principal components, with parameters being $a = 200$, 500, 1000 and $s = 2.5$, 5, 7.5, 10. While d and t remain at their default values.

C. Dataset 3: Yancheng

This dataset comprises bitemporal EO-1 Hyperion images acquired over a wetland agricultural area in Yancheng, China, on May 3, 2005 (X_1) and April 23, 2007 (X_2). The images, with dimensions of 256×91 pixels, contain 132 preprocessed bands, selected out of original 242 bands after preprocessing operations. The dataset comprises four agricultural land-cover change classes, mainly associated with transitions among vegetation, bare land, water, and soil. Fig. 11(a)–(c) illustrates

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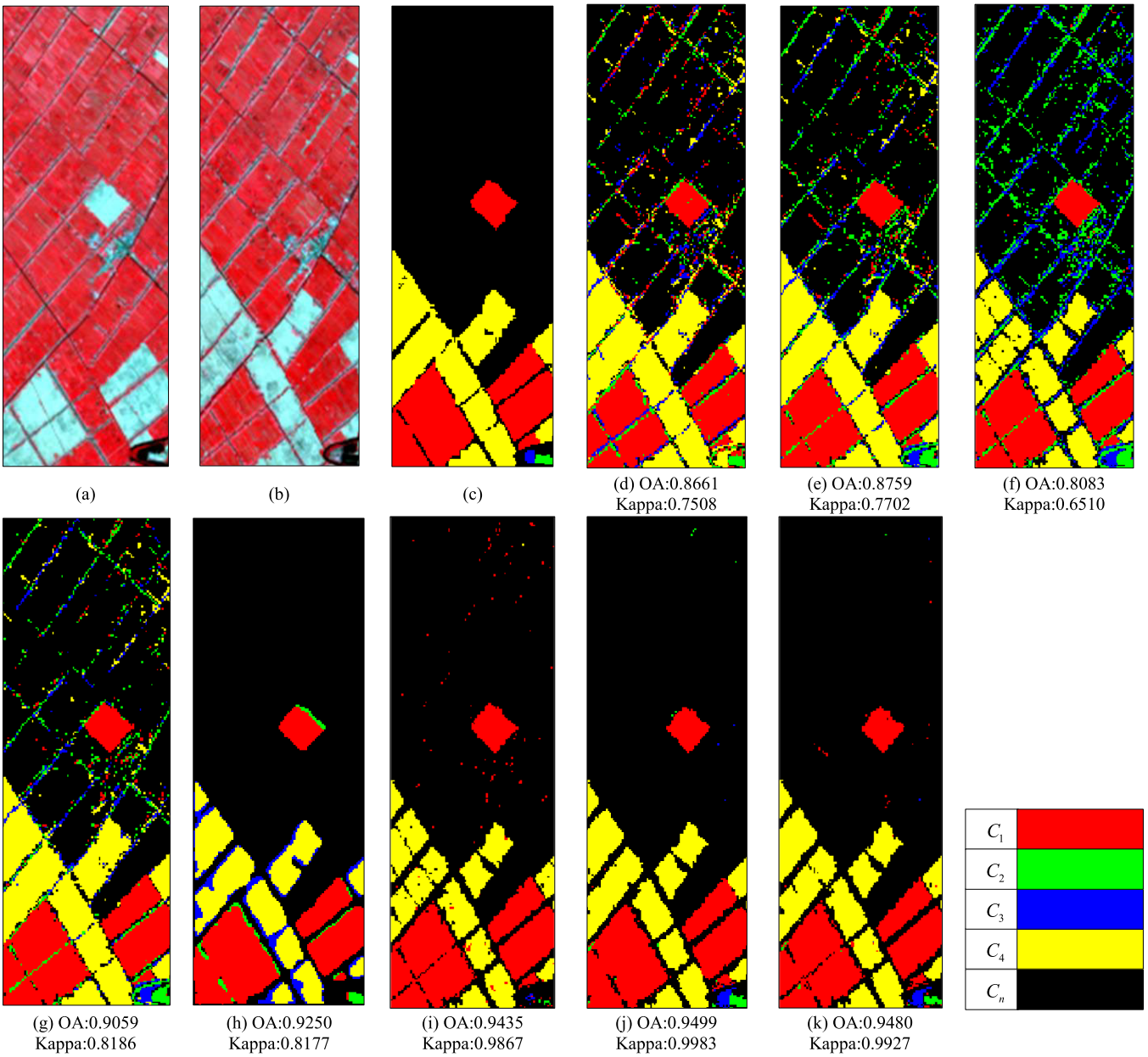


Fig. 11. False-color composites (R: 762.60 nm, G: 640.50 nm, and B: 559.09 nm) of Yancheng dataset (China). (a) Images acquired in 2006 ($\boldsymbol{X}_1$) and (b) images acquired in 2007 ($\boldsymbol{X}_2$). (c) Reference map. (d)–(i) Multiclass CD result obtained by different methods: (d) C^{2}VA, (e) S^{2}CVA, (f) ICVA, (g) S^{2}MCD, (h) KPCA-MNet, and (i) APSG (SVM). (j) APSG (VGG16). (k) APSG_EMAP (SVM).

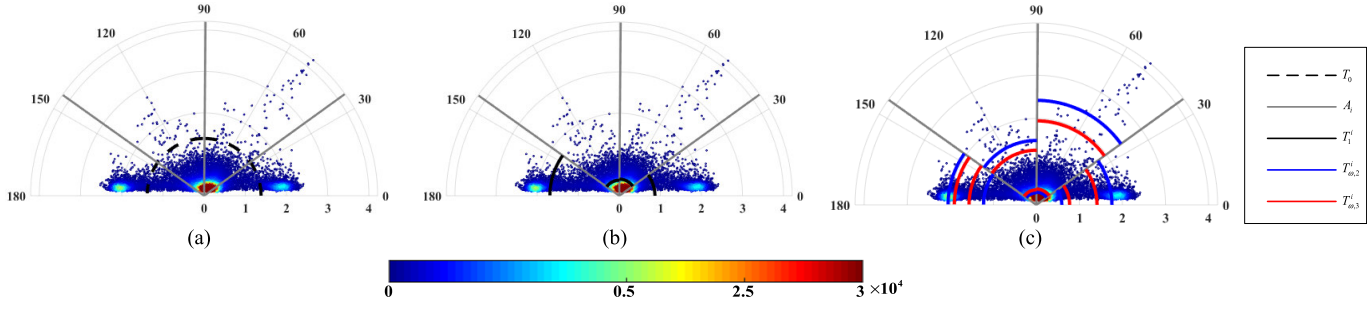


Fig. 12. Pseudo-labeled sample selection region generation results (Yancheng dataset). (a) Sector regions corresponding to each change classes, (b) change and no-change sample region separation in each sector, and (c) final pseudo-labeled sample selection regions.

false-color composites of X_1, X_2, and the reference change map, which categorizes five change classes (i.e., C_1–C_5) and the no-change class (i.e., C_n) within the study area.

The computational process for pseudo-labeled sample generation regions is illustrated in Fig. 12. The semicircular polar coordinate domain is divided into four sectors using threshold

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TABLE III
COMPARISON RESULTS OBTAINED BY DIFFERENT METHODS ON YANCHENG DATASET

Method	OA	Kappa	OE (pixels)	F_1	Time(s)
C^2VA	0.8661	0.7508	3,019	0.8061	6.72
S^2CVA	0.8759	0.7702	2,892	0.8228	**6.01**
ICVA	0.8083	0.6510	4,567	0.7194	10.22
S^2MCD	0.8953±0.0076	0.8088±0.0130	2,432±174	0.8468±0.0091	34.24±11.73
KPCA-MNet	0.9250	0.8177	1747	**0.9610**	52.82
APSG(SVM)	0.9278±0.0251	0.8278±0.2190	1,479±204	0.8849±0.0179	65.51±0.47
APSG(VGG16)	0.9388±0.0062	0.9717±0.0730	**1,368±93**	0.9570±0.0351	152.72±2.53
APSG_EMAP(SVM)	**0.9415±0.0044**	**0.9734±0.0129**	1,424±101	0.8901±0.0074	55.51±3.18

A_i, corresponding to classes C_1–C_4 in a counterclockwise sequence. Change regions associated with C_2 and C_3 are considered susceptible to the long-tail effect.

Numerical experiment results are presented in Table III, with multiclass CD maps shown in Fig. 11(d)–(k). For Yancheng dataset, EMAPs are derived from the first four principal components, with $a = 200$, $d = 100$, $t = 0.5$, and $s = 2.5$. While traditional unsupervised methods detect most changing regions [Fig. 11(d)–(f)], they exhibit limited robustness against noise interference. The S^2MCD method [Fig. 11(g)] suffers from misdetection errors in C_2 and C_3 classes due to inherent bias of single-threshold approaches toward classes with larger sample sizes. KPCA-MNet [Fig. 11(h)] achieved relatively high accuracy and the highest F_1 score on this dataset, owing to the extreme sample imbalance. However, Fig. 11(h) reveals numerous FPs at change region edges, indicating the model's difficulty in accurately detecting samples with weak interclass discrimination. In contrast, the proposed method [Fig. 11(i)–(k)] effectively suppresses background noise and identifies isolated small changes with higher accuracy. Precise delineation of pseudo-labeled sample generation regions ensures proper labeling of the pseudo-labeled sample set, while samples generated from medium-confidence regions enhance its diversity. Both qualitative and quantitative analyses demonstrate the effectiveness of the proposed method, with more than 0.03 increase in OA value [i.e., from 0.8953 ± 0.0076 in S^2MCD to 0.9278 ± 0.0251 in APSG (SVM)]. Furthermore, integration with EMAP features further improves CD performance and achieves the highest accuracy (i.e., OA = 0.9415 ± 0.0044 and Kappa = 0.9734 ± 0.0129).

D. Parameter Analysis

In the proposed APSG approach, the proportion of pseudo-sample generation is a critical hyperparameter that directly influences the OA and time efficiency of the final multiclass CD results. As described in Section III, APSG comprises three sequential pseudo-sample generation steps: rough binary CD, refined binary CD, and multiclass CD. We evaluated the impact of varying pseudo-sample generation proportion on the OA of CD results.

Given the strong correlation between hyperparameters in rough and refined binary CD, these parameters were tested and evaluated jointly. As illustrated in Fig. 13(a)–(c), the pseudo-sample generation proportion for rough binary CD ranges from 2.5% to 17.5% with a step of 2.5%, while for refined binary CD it ranges from 1% to 10% with a 1% step. Across the three datasets, OA remained stable when the proportion of rough binary CD reached 10%. For Yancheng dataset [Fig. 13(c)], OA declined when this proportion reaches 15% or higher due to the introduction of excessive potential mislabeled samples. For refined binary CD, variations in pseudo-sample generation proportion had a negligible impact on APSG's performance for Sardinia dataset [Fig. 13(a)]. However, results for the other two datasets indicate that proportions that are too low or too high significantly reduced the OA of binary CD results, as illustrated in Fig. 13(b) and (c). Therefore, 10% and 5% were selected as the pseudo-sample generation proportions for rough and refined binary CD in APSG, respectively.

As shown in Fig. 13(d) and (e), the OA of multiclass CD results for the three datasets remains stable or gradually declines once the pseudo-sample generation proportion reaches 10%. Similar to binary CD, an excessively high proportion introduces potential mislabeled multiclass samples and substantially increases computational time cost. Consequently, the pseudo-sample generation proportion for multiclass CD is set to 10% to balance time efficiency and OA performance.

V. DISCUSSION

We propose an efficient and effective method for generating pseudo-sample sets in bitemporal remote sensing images to support the multiclass CD task. Compared to traditional unsupervised CD methods and DL-based methods, the proposed APSG technique significantly improves multiclass CD accuracy. By compressing SCVs into a 2-D polar coordinate domain to generate pseudo-samples, APSG achieves broad applicability with few constraints on spatial or spectral resolution. APSG demonstrated high performance across three experimental datasets, including one multispectral and two hyperspectral datasets, with spatial resolutions ranging from 30 to 60 m/pixel. Furthermore, it also exhibits robust CD capabilities in diverse environments, including fire-impacted regions and agricultural areas. Consequently, this approach holds potential for rapid post-disaster assessment and seasonal agricultural monitoring due to its ability to deliver multiclass CD results without prior information.

However, the effectiveness of APSG is limited by the complexity of CD scenarios. As the method visualizes data by compressing high-dimensional SCVs into two dimensions, spectral feature information is inevitably lost. Therefore, change classes exhibiting minor spectral variations may not be significantly distinguishable in the polar coordinate domain after compression. Furthermore, the distribution of different change classes within this domain relies on thresholds applied

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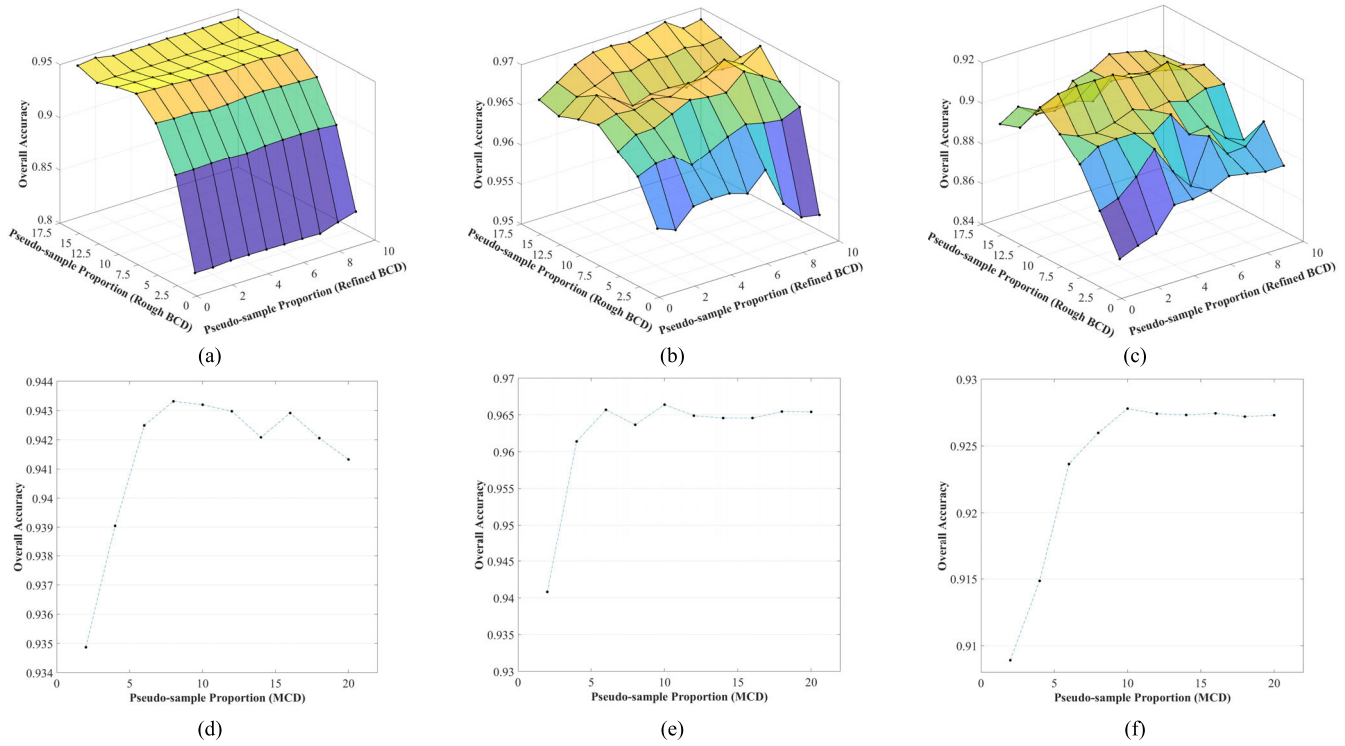


Fig. 13. Comparison of CD results obtained on different proportions of pseudo-sample training sets. (a)–(c) Binary CD performances based on different pseudo-sample proportions for rough and refined binary CD stages for Sardinia, Umatilla, and Yancheng datasets, respectively. (d)–(f) Multiclass CD performances on Sardinia, Umatilla, and Yancheng datasets based on different multiclass pseudo-sample proportions, respectively.

to the direction θ, which also limits the complexity of scenarios APSG can handle. When a scene contains an excessive number of change classes (e.g., more than ten change classes), accurately separating each change class within the 180° polar coordinate domain using thresholds becomes difficult. This may introduce a significant number of mislabeled training samples during the pseudo-sample generation step in APSG. Similar to the framework proposed in [33], APSG supports iterative refinement in such scenarios.

VI. Conclusion

This article has proposed the novel APSG technique that generates pseudo-labeled sample training sets for multiclass CD in bitemporal remote sensing images. As an unsupervised approach, APSG eliminates the requirement for prior ground reference data. The main contributions are summarized as follows.

1) A novel adaptive pseudo-labeled sample generation (APSG) approach is proposed to automatically delineate candidate regions for pseudo-labeled sample selection with distinct confidence levels and generate the pseudo-labeled sample training sets in an unsupervised manner.

2) A sequential pseudo-labeled sample generation strategy, incorporated with the transition from binary to multiclass CD, is designed based on APSG to enhance sample diversity and mitigate the impact of mislabeled samples on CD accuracy.

3) Spectral–spatial change information is integrated into the proposed approach to improve the representativeness and stability of the pseudo-labeled sample training set, particularly for isolated small changes, thereby enhancing CD accuracy.

Compared with magnitude-based methods using a single threshold, the proposed APSG approach generates higher precision training sets and yields more accurate CD results. Experimental validation on three bitemporal remote sensing datasets with varying spatial and spectral resolution confirms the effectiveness of the proposed approach.

For future work, we will focus on enhancing spectral discriminability for subtle changes and refining pseudo-labeled samples to further improve CD accuracy. In addition, we plan to integrate the pseudo-sample generation approach with state-of-the-art DL models to further investigate APSG's potential in applications such as rapid disaster area monitoring and the analysis of seasonal variations in agricultural land cover.

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